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MCS-012

**MASTER OF COMPUTER
APPLICATION/BACHELOR OF
COMPUTER APPLICATION
(REVISED) (MCA/BCA)
Term-End Examination**

June, 2020

**MCS-012 : COMPUTER ORGANIZATION AND
ASSEMBLY LANGUAGE PROGRAMMING**

Time : 3 Hours

Maximum Marks : 100

Weighthage : 75%

**Note : (i) Question No. 1 is compulsory and carries
40 marks.**

**(ii) Attempt any three questions from the
rest.**

1. (a) Convert the following numbers as stated : 6

(i) $(23.125)_{10}$ to binary

(ii) $(36.5)_{10}$ to octal

(iii) $(135)_{10}$ to hexadecimal

(b) Draw the truth table for the following Boolean function :

$$F = (A \cdot B + C) + (\bar{A} \cdot C) + (\bar{B} \cdot A \cdot \bar{C})$$

Use k-map to simplify the above Boolean function. 5

(c) What is the need of Cache memory ?

Explain the direct Cache mapping scheme with the help of an example/diagram. 6

(d) Explain the interrupt-driven I/O technique with the help of a diagram. 4

(e) How is the next instruction that is to be executed brought into Instruction Register for execution ? Explain the sequence of micro-operation that are needed to perform this operation. Which of these micro-operations will take longest time to execute ? Give justification in support of your answer. 6

(f) How is the large register file of RISC useful ? 3

(g) Write a program using 8086 assembly language that finds the larger of two byte values stored in two memory locations named A and B respectively. The larger of two values should be stored in AL register. 6

- (h) Calculate the physical address for the following segment register : 4

Offset (or Register) pair

- (i) Offset of data byte in segment $(0200)_h$

Data Segment (DS) : $(1FFF)_h$

- (ii) Code Segment Register (CS) : $0F10$

Instruction Pointer Register (IP) :

2562

2. (a) Explain the 'Stored Program Concept' for a Von Neumann machine. 4

- (b) What is an I/O processor ? Explain its characteristics. Explain the selector and multiplexer channels with the help of diagram(s). 6

- (c) Consider the Registers R1 having value $(1011\ 0101)_2$ and R2 having value

$(0110\ 0111)_2$. Perform the following operations using R1 and/or R2. The result should be stored in a register R :

(i) Addition of R1 and R2 with carry

(ii) Decrement R1

(iii) Increment R1

(iv) Subtract R2 from R1

It may be noted that only addition micro-operation is allowed. 6

(d) Explain the FAR procedure call in the context of 8086 assembly language. 4

3. (a) What is an Interrupt Vector Table in 8086 microprocessor ? How is it used to process an Interrupt ? Explain with the help of a diagram. 6

- (b) Explain the following in the context of micro-programmed control unit : 6
- (i) Control memory
 - (ii) Sequencing logic
 - (iii) Vertical micro-instruction
- (c) Explain the following in the context of printing technology : 6
- (i) Print quality
 - (ii) Impact and non-impact printers
 - (iii) Print resolution
- (d) List any *four* advantages of densely packed integrated circuits. 2
4. (a) What is the use of Multiplexers ? Draw and explain the logic diagram of a 4×1 multiplexer. Also draw the truth table for this multiplexer. 6

(b) A memory chip has a capacity of $1 \text{ M} \times 16$ bits : 4

(i) How many address lines does it have ?

(ii) What is the capacity of the chip in bytes ?

(c) What is an Accumulator base Instruction Set Architecture ? Write the assembly code for the expression $A = B * C + D$ for Accumulator based machine. 6

(d) What is the role of Flag register in 8086 microprocessor ? Explain the role of any *three* flags in this register. 4

5. Explain briefly any *eight* of the following :

$$8 \times 2\frac{1}{2} = 20$$

(a) Assembler

(b) Stack segment

- (c) EXE programs
- (d) Shift instruction
- (e) D flip-flop
- (f) Memory interleaving
- (g) Latency time in disk access
- (h) Normalization of floating point numbers
- (i) Unicode
- (j) Counters